

DMH 25x, DMH 28x, DMX 226

ATEX-approved pumps

Installation and operating instructions, supplement



Declaration of conformity

GB: EC declaration of conformity

We, Grundfos, declare under our sole responsibility that the products DMH 25x...E1/E2, DMH 28x...E1/E2, DMX 226...E1/E2, to which this declaration relates, are in conformity with these Council directives on the approximation of the laws of the EC member states:

- ATEX 94/9/EC (ATEX 100).
Standards used: EN 1127-1:1997, EN 13463-1:2001, EN 13463-5:2003.
- Machinery Directive (2006/42/EC).
Standards used: EN 809:1998+A1:2009, EN ISO 12100-1+A1:2009, EN ISO 12100-2+A1:2009.

Additionally for the pump as a unit with motor:

- Low Voltage Directive (2006/95/EC).
Standard used: EN 61010-1:2001 (second edition).
- EMC Directive (2004/108/EC).*

* Only for products with control variant AT.

This EC declaration of conformity is only valid when published as part of the Grundfos installation and operating instructions.

DE: EG-Konformitätserklärung

Wir, Grundfos, erklären in alleiniger Verantwortung, dass die Produkte DMH 25x...E1/E2, DMH 28x...E1/E2, DMX 226...E1/E2, auf die sich diese Erklärung bezieht, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EU-Mitgliedsstaaten übereinstimmen:

- ATEX 94/9/EC (ATEX 100).
Normen, die verwendet wurden: EN 1127-1:1997, EN 13463-1:2001, EN 13463-5:2003.
- Maschinenrichtlinie (2006/42/EG).
Normen, die verwendet wurden: EN 809:1998+A1:2009, EN ISO 12100-1+A1:2009, EN ISO 12100-2+A1:2009.

Zusätzlich für die Pumpe als Gerät mit Motor:

- Niederspannungsrichtlinie (2006/95/EG).
Norm, die verwendet wurde: EN 61010-1:2001 (zweite Ausgabe).
- EMV-Richtlinie (2004/108/EG).*

* Nur für Produkte mit Steuerungsvariante AT.

Diese EG-Konformitätserklärung gilt nur, wenn sie in Verbindung mit der Grundfos Montage- und Betriebsanleitung veröffentlicht wird.

ES: Declaración CE de conformidad

Nosotros, Grundfos, declaramos bajo nuestra entera responsabilidad que los productos DMH 25x...E1/E2, DMH 28x...E1/E2, DMX 226...E1/E2, a los cuales se refiere esta declaración, están conformes con las Directivas del Consejo en la aproximación de las leyes de los Estados Miembros del EM:

- ATEX 94/9/EC (ATEX 100).
Normas aplicadas: EN 1127-1:1997, EN 13463-1:2001, EN 13463-5:2003.
- Directiva de Maquinaria (2006/42/CE).
Normas aplicadas: EN 809:1998+A1:2009, EN ISO 12100-1+A1:2009, EN ISO 12100-2+A1:2009.

Adicionalmente para la bomba como una unidad con motor:

- Directiva de Baja Tensión (2006/95/CE).
Norma aplicada: EN 61010-1:2001 (segunda edición).
- Directiva EMC (2004/108/CE).*

* Solo para productos con variante de control AT.

Esta declaración CE de conformidad sólo es válida cuando se publique como parte de las instrucciones de instalación y funcionamiento de Grundfos.

CZ: ES prohlášení o shodě

My firma Grundfos prohlašujeme na svou plnou odpovědnost, že výrobky DMH 25x...E1/E2, DMH 28x...E1/E2, DMX 226...E1/E2, na něž se toto prohlášení vztahuje, jsou v souladu s ustanoveními směrnice Rady pro sblížení právních předpisů členských států Evropského společenství v oblastech:

- ATEX 94/9/EC (ATEX 100).
Použité normy: EN 1127-1:1997, EN 13463-1:2001, EN 13463-5:2003.
- Směrnice pro strojní zařízení (2006/42/ES).
Použité normy: EN 809:1998+A1:2009, EN ISO 12100-1+A1:2009, EN ISO 12100-2+A1:2009.

Navíc pro čerpadla jako jednotky s motorem:

- Směrnice pro nízkonapěťové aplikace (2006/95/ES).
Použitá norma: EN 61010-1:2001 (druhé vydání).
- Směrnice pro elektromagnetickou kompatibilitu (EMC) (2004/108/ES).*

* Pouze pro výrobky s variantou řízení AT.

Toto ES prohlášení o shodě je platné pouze tehdy, pokud je zveřejněno jako součást instalačních a provozních návodů Grundfos.

GR: Δήλωση συμμόρφωσης EC

Εμείς, η Grundfos, δηλώνουμε με αποκλειστικά δική μας ευθύνη ότι τα προϊόντα DMH 25x...E1/E2, DMH 28x...E1/E2, DMX 226...E1/E2 στα οποία αναφέρεται η παρούσα δήλωση, συμμορφώνονται με τις εξής Οδηγίες του Συμβουλίου περί προσέγγισης των νομοθεσιών των κρατών μελών της ΕΕ:

- ATEX 94/9/EC (ATEX 100).
Πρότυπα που χρησιμοποιήθηκαν: EN 1127-1:1997, EN 13463-1:2001, EN 13463-5:2003.
- Οδηγία για μηχανήματα (2006/42/ΕC).
Πρότυπα που χρησιμοποιήθηκαν: EN 809:1998+A1:2009, EN ISO 12100-1+A1:2009, EN ISO 12100-2+A1:2009.

Επιπρόσθετα για την αντλία ως μονάδα με κινητήρα:

- Οδηγία χαμηλής τάσης (2006/95/ΕC).
Πρότυπο που χρησιμοποιήθηκε: EN 61010-1:2001 (δεύτερη έκδοση).
- Οδηγία Ηλεκτρομαγνητικής Συμβατότητας (EMC) (2004/108/ΕC).*

* Μόνο για προϊόντα με τύπο ελέγχου AT.

Αυτή η δήλωση συμμόρφωσης EC ισχύει μόνον όταν συνοδεύει τις οδηγίες εγκατάστασης και λειτουργίας της Grundfos.

FR: Déclaration de conformité CE

Nous, Grundfos, déclarons sous notre seule responsabilité, que les produits DMH 25x...E1/E2, DMH 28x...E1/E2, DMX 226...E1/E2, auxquels se réfère cette déclaration, sont conformes aux Directives du Conseil concernant le rapprochement des législations des Etats membres CE relatives aux normes énoncées ci-dessous :

- ATEX 94/9/EC (ATEX 100).
Normes utilisées : EN 1127-1:1997, EN 13463-1:2001, EN 13463-5:2003.
- Directive Machines (2006/42/CE).
Normes utilisées : EN 809:1998+A1:2009, EN ISO 12100-1+A1:2009, EN ISO 12100-2+A1:2009.

Également pour la pompe comme unité avec moteur :

- Directive Basse Tension (2006/95/CE).
Norme utilisée : EN 61010-1:2001 (deuxième édition).
- Directive Compatibilité Electromagnétique CEM (2004/108/CE).*

* Uniquement pour produits avec variante de commande AT.

Cette déclaration de conformité CE est uniquement valide lors de sa publication dans la notice d'installation et de fonctionnement Grundfos.

HR: EZ izjava o usklađenosti

Mi, Grundfos, izjavljujemo pod vlastitom odgovornošću da je proizvod DMH 25x...E1/E2, DMH 28x...E1/E2, DMX 226...E1/E2, na koji se ova izjava odnosi, u skladu s direktivama ovog Vijeća o usklađivanju zakona država članica EU:

- ATEX 94/9/EC (ATEX 100).
Korištene norme: EN 1127-1:1997, EN 13463-1:2001, EN 13463-5:2003.
- Direktiva za strojeve (2006/42/EZ).
Korištene norme: EN 809:1998+A1:2009, EN ISO 12100-1+A1:2009, EN ISO 12100-2+A1:2009.

Dodatno za crpku kao jedinicu s motorom:

- Direktiva za niski napon (2006/95/EZ).
Korištena norma: EN 61010-1:2001 (drugo izdanje).
- Direktiva za elektromagnetsku kompatibilnost (2004/108/EZ).*

* Samo za proizvode s vrstom upravljanja AT.

Ova EZ izjava o usklađenosti važeća je jedino kada je izdana kao dio Grundfos montažnih i pogonskih uputa.

HU: EK megfelelősségi nyilatkozat

Mi, a Grundfos, egyedüli felelősséggel kijelentjük, hogy a DMH 25x...E1/E2, DMH 28x...E1/E2, DMX 226...E1/E2 termékek, amelyekre jelen nyilatkozik vonatkozik, megfelelnek az Európai Unió tagállamainak jogi irányelveit összehangoló tanács alábbi előírásainak:

- ATEX 94/9/EC (ATEX 100).
Alkalmazott szabványok: EN 1127-1:1997, EN 13463-1:2001, EN 13463-5:2003.
- Gépek (2006/42/EK).
Alkalmazott szabványok: EN 809:1998+A1:2009, EN ISO 12100-1+A1:2009, EN ISO 12100-2+A1:2009.

Kiegészítés, szivattyút és motort egy egységnek tekintve:

- Kisfeszültségű Direktiva (2006/95/EK).
Alkalmazott szabvány: EN 61010-1:2001 (második kiadás).
- EMC Direktiva (2004/108/EK).*

* Csak AT vezérlési változat esetén.

Ez az EK megfelelőségi nyilatkozat kizárólag akkor érvényes, ha Grundfos telepítési és üzemeltetési utasítás részeként kerül kiadásra.

PL: Deklaracja zgodności WE

My, Grundfos, oświadczamy z pełną odpowiedzialnością, że nasze wyroby DMH 25x...E1/E2, DMH 28x...E1/E2, DMX 226...E1/E2, których deklaracja niniejsza dotyczy, są zgodne z następującymi wytycznymi Rady d/s ujednolicenia przepisów prawnych krajów członkowskich WE:

- ATEX 94/9/EC (ATEX 100).
Zastosowane normy: EN 1127-1:1997, EN 13463-1:2001, EN 13463-5:2003.
- Dyrektywa Maszynowa (2006/42/WE).
Zastosowane normy: EN 809:1998+A1:2009, EN ISO 12100-1+A1:2009, EN ISO 12100-2+A1:2009.

Dodatkowo dla pompy jako zespołu z silnikiem:

- Dyrektywa Niskonapięciowa (LVD) (2006/95/WE).
Zastosowana norma: EN 61010-1:2001 (drugie wydanie).
- Dyrektywa EMC (2004/108/WE).*

* Tylko dla produktów w wersji sterowania AT.

Deklaracja zgodności WE jest ważna tylko i wyłącznie wtedy kiedy jest opublikowana przez firmę Grundfos i umieszczona w instrukcji montażu i eksploatacji.

RU: Декларация о соответствии ЕС

Мы, компания Grundfos, со всей ответственностью заявляем, что изделия DMH 25x...E1/E2, DMH 28x...E1/E2, DMX 226...E1/E2, к которым относится настоящая декларация, соответствуют следующим Директивам Совета Евросоюза об унификации законодательных предписаний стран-членов ЕС:

- ATEX 94/9/EC (ATEX 100).
Применявшиеся стандарты: EN 1127-1:1997, EN 13463-1:2001, EN 13463-5:2003.
- Механические устройства (2006/42/ЕС).
Применявшиеся стандарты: EN 809:1998+A1:2009, EN ISO 12100-1+A1:2009, EN ISO 12100-2+A1:2009.

Дополнительно для насоса в качестве устройства с двигателем:

- Низковольтное оборудование (2006/95/ЕС).
Применявшийся стандарт: EN 61010-1:2001 (второе издание).
- Электромагнитная совместимость (2004/108/ЕС).*

* Только для насосов с блоком управления AT.

Данная декларация о соответствии ЕС имеет силу только в случае публикации в составе инструкции по монтажу и эксплуатации на продукцию производства компании Grundfos.

IT: Dichiarazione di conformità CE

Grundfos dichiara sotto la sua esclusiva responsabilità che i prodotti DMH 25x...E1/E2, DMH 28x...E1/E2, DMX 226...E1/E2, ai quali si riferisce questa dichiarazione, sono conformi alle seguenti direttive del Consiglio riguardanti il riavvicinamento delle legislazioni degli Stati membri CE:

- ATEX 94/9/EC (ATEX 100).
Norme applicate: EN 1127-1:1997, EN 13463-1:2001, EN 13463-5:2003.
- Direttiva Macchine (2006/42/CE).
Norme applicate: EN 809:1998+A1:2009, EN ISO 12100-1+A1:2009, EN ISO 12100-2+A1:2009.

In aggiunta, per la pompa come unità con motore:

- Direttiva Bassa Tensione (2006/95/CE).
Norma applicata: EN 61010-1:2001 (seconda edizione).
- Direttiva EMC (2004/108/CE).*

* Solo per prodotti con varianti di controllo AT.

Questa dichiarazione di conformità CE è valida solo quando pubblicata come parte delle istruzioni di installazione e funzionamento Grundfos.

NL: EC overeenkomstigheidsverklaring

Wij, Grundfos, verklaren geheel onder eigen verantwoordelijkheid dat de producten DMH 25x...E1/E2, DMH 28x...E1/E2, DMX 226...E1/E2 waarop deze verklaring betrekking heeft, in overeenstemming zijn met de Richtlijnen van de Raad in zake de onderlinge aanpassing van de wetgeving van de EG Lidstaten betreffende:

- ATEX 94/9/EC (ATEX 100).
Gebruikte normen: EN 1127-1:1997, EN 13463-1:2001, EN 13463-5:2003.
- Machine Richtlijn (2006/42/EC).
Gebruikte normen: EN 809:1998+A1:2009, EN ISO 12100-1+A1:2009, EN ISO 12100-2+A1:2009.

Aanvullend voor de pomp als een eenheid met motor:

- Laagspannings Richtlijn (2006/95/EC).
Gebruikte norm: EN 61010-1:2001 (tweede editie).
- EMC Richtlijn (2004/108/EC).*

* Enkel voor producten met besturingsvariant AT.

Deze EC overeenkomstigheidsverklaring is alleen geldig wanneer deze gepubliceerd is als onderdeel van de Grundfos installatie- en bedieningsinstructies.

PT: Declaração de conformidade CE

A Grundfos declara sob sua única responsabilidade que os produtos DMH 25x...E1/E2, DMH 28x...E1/E2, DMX 226...E1/E2, aos quais diz respeito esta declaração, estão em conformidade com as seguintes Directivas do Conselho sobre a aproximação das legislações dos Estados Membros da CE:

- ATEX 94/9/EC (ATEX 100).
Normas utilizadas: EN 1127-1:1997, EN 13463-1:2001, EN 13463-5:2003.
- Directiva Máquinas (2006/42/CE).
Normas utilizadas: EN 809:1998+A1:2009, EN ISO 12100-1+A1:2009, EN ISO 12100-2+A1:2009.

Adicionalmente para a bomba como uma unidade com o motor:

- Directiva Baixa Tensão (2006/95/CE).
Norma utilizada: EN 61010-1:2001 (segunda edição).
- Directiva EMC (compatibilidade electromagnética) (2004/108/CE).*

* Apenas para produtos com sistema de controlo AT.

Esta declaração de conformidade CE é apenas válida quando publicada como parte das instruções de instalação e funcionamento Grundfos.

SI: ES izjava o skladnosti

V Grundfosu s polno odgovornostjo izjavljamo, da so naši izdelki DMH 25x...E1/E2, DMH 28x...E1/E2, DMX 226...E1/E2, na katere se ta izjava nanaša, v skladu z naslednjimi direktivami Sveta o približevanju zakonodaje za izenačevanje pravnih predpisov držav članic ES:

- ATEX 94/9/EC (ATEX 100).
Uporabljeni normi: EN 1127-1:1997, EN 13463-1:2001, EN 13463-5:2003.
- Direktiva o strojih (2006/42/ES).
Uporabljeni normi: EN 809:1998+A1:2009, EN ISO 12100-1+A1:2009, EN ISO 12100-2+A1:2009.

Dodatno za črpalko kot enoto z motorjem:

- Direktiva o nizki napetosti (2006/95/ES).
Uporabljen normi: EN 61010-1:2001 (druga izdaja).
- Direktiva o elektromagnetni združljivosti (EMC) (2004/108/ES).*

* Samo za izdelke z kontrolno različico AT.

ES izjava o skladnosti velja samo kadar je izdana kot del Grundfos instalacije in navodil delovanja.

SE: EG-försäkran om överensstämmelse

Vi, Grundfos, försäkrar under ansvar att produkterna DMH 25x...E1/E2, DMH 28x...E1/E2, DMX 226...E1/E2, som omfattas av denna försäkran, är i överensstämmelse med rådets direktiv om inbördes närmande till EU-medlemsstaternas lagstiftning, avseende:

- ATEX 94/9/EC (ATEX 100).
Tillämpade standarder: EN 1127-1:1997, EN 13463-1:2001, EN 13463-5:2003.
- Maskindirektivet (2006/42/EG).
Tillämpade standarder: EN 809:1998+A1:2009, EN ISO 12100-1+A1:2009, EN ISO 12100-2+A1:2009.

Dessutom för pumpen såsom en enhet med motor:

- Lågspänningsdirektivet (2006/95/EG).
Tillämpad standard: EN 61010-1:2001 (andra upplagan).
- EMC-direktivet (2004/108/EG).*

* Endast för produkter med styrenhetsvariant AT.

Denna EG-försäkran om överensstämmelse är endast giltig när den publiceras som en del av Grundfos monterings- och driftsinstruktion.

CN: EC 产品合格声明书

我们格兰富在我们的全权责任下声明，产品 DMH 25x...E1/E2, DMH 28x...E1/E2, DMX 226...E1/E2, 即该合格证所指之产品，符合欧共体使其成员国法律趋于一致的以下欧共理事会指令：

- ATEX 94/9/EC (ATEX 100).
所用标准：EN 1127-1:1997, EN 13463-1:2001, EN 13463-5:2003。
- 机械设备指令 (2006/42/EC)。
所用标准：EN 809:1998+A1:2009, EN ISO 12100-1+A1:2009, EN ISO 12100-2+A1:2009。

对于带电机的泵总成设备的附加标准：

- 低电压指令 (2006/95/EC)。
所用标准：EN 61010-1:2001 (第 2 版)。
- 电磁兼容性指令 (2004/108/EC)。*

* 仅对控制选项为 AT 的型号有效。

本 EC 合格性声明仅在作为格兰富安装与操作指导手册的一部分时有效。

TR: EC uygunluk bildirgesi

Grundfos olarak bu beyannameye konu olan DMH 25x...E1/E2, DMH 28x...E1/E2, DMX 226...E1/E2 ürünlerinin, AB Üyesi Ülkelerin kanunlarını birbirine yaklaştırma üzerine Konsey Direktifleriyle uyumlu olduğunun yalnızca bizim sorumluluğumuz altında olduğunu beyan ederiz:

- ATEX 94/9/EC (ATEX 100).
Kullanılan standartlar: EN 1127-1:1997, EN 13463-1:2001, EN 13463-5:2003.
- Makineler Yönetmeliği (2006/42/EC).
Kullanılan standartlar: EN 809:1998+A1:2009, EN ISO 12100-1+A1:2009, EN ISO 12100-2+A1:2009.

Ayrıca, motorlu bir ünite olarak pompa için:

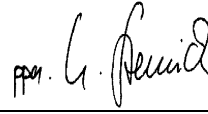
- Düşük Voltaj Yönetmeliği (2006/95/EC).
Kullanılan standart: EN 61010-1:2001 (ikinci baskı).
- EMC Direktifi (2004/108/EC).*

* Sadece AT kontrol değişkenli ürünler için.

İşbu EC uygunluk bildirgesi, yalnızca Grundfos kurulum ve çalıştırma talimatlarının bir parçası olarak basıldığı takdirde geçerlilik kazanmaktadır.

EC-type examination certificate number: IBExU06ATEXB012 X
Notified body number: 0637 (IBExU Institut für Sicherheitstechnik GmbH)

Pfintzal, 15th May, 2012



Ulrich Stemick
Technical Director
Grundfos Water Treatment GmbH
Reetzstr. 85, D-76327 Pfintzal, Germany

Person authorised to compile technical file and
empowered to sign the EC declaration of conformity

Original installation and operating instructions.

CONTENTS

	Page
1. Symbols used in this document	5
2. Operating pumps in potentially explosive atmospheres	5
3. Device designs	5
3.1 Identification	5
3.2 Pump configuration	5
4. Operating conditions	7
4.1 Environmental conditions	7
4.2 Dosing medium	7
5. Installation	7
5.1 Earthing	7
5.2 Overflow valve	7
5.3 Suction line	7
5.4 Dosing of combustible media	8
6. Commissioning	8
7. Operation	8
7.1 Safety instructions for operating the explosion-proof pump	8
8. Maintenance	9
8.1 Cleaning	9
8.2 Lubrication	9
8.3 Accessories / peripherals	9
8.4 Tools and maintenance work	9
9. Declaration of conformity / certificates	9

**Warning**

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

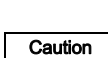
**Warning**

In addition to these instructions, observe the general installation and operating instructions supplied with the pump.

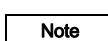
1. Symbols used in this document

**Warning**

If these safety instructions are not observed, it may result in personal injury.



If these safety instructions are not observed, it may result in malfunction or damage to the equipment.



Notes or instructions that make the job easier and ensure safe operation.

2. Operating pumps in potentially explosive atmospheres

This manual describes additional conditions or special considerations affecting the operation of pumps in potentially explosive atmospheres.



This manual is only valid together with the installation and operating instructions for the pump. Both documents should be used.



Installation, maintenance and repairs must only be carried out by suitably qualified and authorised personnel.

3. Device designs

Caution *The motor and accessories for the pumps, e.g. the metrology-based monitoring, must at minimum meet the requirements of Directive 94/9/EC as applicable to pumps.*

Caution *The original configuration of the pump must not be modified.*

3.1 Identification

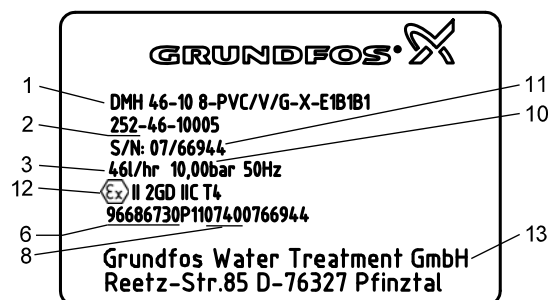


Fig. 1 Example of ATEX nameplate

Pos.	Description
1	Type designation
2	Model
3	Maximum capacity [l/h]
6	Product number
8	Year and week code
10	Maximum pressure [bar]
11	Serial number
12	ATEX designation
13	Manufacturer

3.2 Pump configuration

Grundfos produces pumps in accordance with Directive 94/9/EC. These pumps correspond to the specifications in the table below.

Caution *Pumps which are not configured in accordance with the table cannot be used in potentially explosive atmospheres.*

Permissible pump configuration

Feature	Description
Front plate	Metallic
Electronics	No electronics

3.2.1 Diaphragm leakage detection

DMH pumps meeting the requirements of Directive 94/9/EC can optionally be fitted with an approved diaphragm leakage detection with an Ex i pressure gauge meeting the requirements of Directive 94/9/EC.

Caution *Pumps with other types of diaphragm leakage detection cannot be used in potentially explosive atmospheres.*

TM03 7006 2011

3.2.2 Dosing head and valves

Grundfos supplies pumps meeting the requirements of Directive 94/9/EC with appropriately approved dosing head/valve combinations meeting the requirements of Directive 94/9/EC.

Caution *Pumps with other types of dosing head/valve combination cannot be used in potentially explosive atmospheres.*

Caution *Stainless-steel valves must not be combined with non-metal dosing heads.*

The size of the dosing head (depending on the pump type) and the dosing head material determine the suitability of the pump for operation in potentially explosive atmospheres caused by media of group IIB or IIC:

Dosing head material	Pump types				
	DMH 251-255	DMH 257	DMH 28x	DMX 226 Head sizes 1-3	DMX 226 Head size 4
PVC	IIC	IIB	IIC	IIC	IIB
Stainless steel, DIN 1.4571	IIC	IIC	IIC	IIC	IIC
Stainless steel, DIN 2.4610	IIC	IIC	IIC	-	-
PP	IIC	IIB	IIC	IIC	IIB
PVDF	IIC	IIB	IIC	IIC	IIB

Nameplate of pump with motor

Motor nameplate	Pump nameplate	Suitable for category 2, explosion group / temperature class
 II 2G Ex e II T3	 II 2G IIB T3	IIB / T3
 II 2G Ex e II T3	 II 2G IIC T3	IIC / T3
 II 2GD Ex die IIC T4	 II 2GD IIB T4	IIB / T4
 II 2GD Ex die IIC T4	 II 2GD IIC T4	IIC / T4

Further information about using the pump can be found in section 4. *Operating conditions.*

3.2.3 Motor

Grundfos pumps complying with Directive 94/9/EC are supplied with motors that are complying with Directive 94/9/EC and have been tested and labelled accordingly.

Alternatively, pump units can be supplied separately, and in this case the user must fit a motor that complies with Directive 94/9/EC.

Caution *Pumps with other types of motor cannot be used in potentially explosive atmospheres.*

3.2.4 Pump designation

The information meeting the requirements of Directive 94/9/EC given on the pump as a non-electrical device depends on the pump/motor configuration selected.

Pumps with motor are labelled in accordance with Directive 94/9/EC. However, if the pump is supplied without a motor, the labelling is only applicable to the pump unit.

Nameplate of the pump unit only (pump supplied without motor)

Pump unit nameplate	Suitable for category 2, explosion group / temperature class
 II 2GD c IIB T130 °C	IIB / 130 °C
 II 2GD c IIC T130 °C	IIC / 130 °C

4. Operating conditions

4.1 Environmental conditions

The pumps with appropriate nameplates are suitable for operation in potentially explosive atmospheres Zones IIB or IIC, depending on the type of pump and its equipment.

Minimum/maximum ambient temperature T_{amb} .

- Minimum T_{amb} : -20 °C.
- Maximum T_{amb} : +40 °C.

The materials used for seals and lubricants must be resistant to vapours, gases or dust generated in the atmosphere of the operating area. This applies particularly to contact with solvent vapours which can damage plastics or change the composition of oil or gear lubricant, thereby affecting its efficiency/ability to function.

The pumps may only be used if, under the prevailing operating conditions, their materials and operating liquids are sufficiently resistant to mechanical and/or chemical influences in order not to jeopardise explosion protection.

Caution

4.2 Dosing medium

4.2.1 Permissible temperature

For DMH 28x pumps, the permissible media temperature deviates from the information in the general installation and operating instructions.

Note

Permissible media temperature

Dosing head material	Max. media temperature			
	p < 10 bar		p < 16 bar	
	Min. [°C]	Max. [°C]	Min. [°C]	Max. [°C]
PVC	0	40	0	20
Stainless steel, DIN 1.4571	-10	70 *	-10	70 *
Stainless steel, DIN 2.4610	-10	70 *	-10	70 *
PP	0	40	-	-
PVDF *	-10	60	-10	20
* (< 9 bar: 70)				

* 90 °C with DMH

4.2.2 Permissible dosing media

The DMH 25x, DMH 28x and DMX 226 pumps are suitable for liquid, non-abrasive and non-inflammable media.

The DMH 25x and DMH 28x pumps in the following configuration can also be used for dosing combustible media:

- Explosion-proof design with appropriate designation on the pump nameplate, see section 3.1 Identification, and
- stainless-steel dosing heads and valves, and
- stainless-steel design of the diaphragm leakage detection in the explosion-proof design with evaluator.

5. Installation

The operator is responsible for setting up the system in accordance with the requirements of Directive 99/92/EC, taking into account the appropriate standards for its application.

The motor and accessories for the pumps, e.g. the actuator, frequency converter and metrology-based monitoring, must at minimum meet the requirements of Directive 94/9/EC as applicable to pumps. Accessories for the system and pipeline must at minimum meet the requirements of Directive 94/9/EC as applicable to pumps.

Caution

An intrinsically safe installation must only be made and checked by someone who is "appropriately qualified".

Caution

Make sure you read the installation and operating instructions supplied with the individual electrical components.

Make sure you adhere to applicable national regulations when installing the system.

5.1 Earthing

Caution

All pumps and accessories for the installation must be earthed.

The plastic version of the diaphragm leakage detection does not have a conductive connection to the pump and must be earthed by the customer.

5.2 Overflow valve

Caution

DMX 226: An overflow valve must be installed in the discharge line to prevent impermissibly high pressure on the discharge side of the pump.

DMH 25x and DMH 28x: If the DMH 25x or DMH 28x is the only pressure source in the system, it is not necessary to install an overflow valve in the discharge line.

5.3 Suction line

Impurities in the dosing medium may damage the diaphragm or block the discharge side of the dosing head.

Caution

Install a filter (aperture size 1.5 mm) in the suction line to prevent the pressure increasing in the event of a fault due to a blockage on the discharge side and damage to the diaphragm.

5.4 Dosing of combustible media

For dosing combustible media, see section 4.2.2 *Permissible dosing media*.

5.4.1 Installation for feed mode



Warning

The pump must not run dry, even during commissioning/recommissioning.

The pump should be installed so that the discharge line can be deaerated during start-up of the pump to enable the dosing medium to flow into the dosing head.

5.4.2 Draining off the dosing medium

Lines for draining off media safely should be connected to the safety valves (such as the overflow valve in the discharge line on the discharge side of the pump) or vent valves via which the combustible media are drained off.

Caution

The media must be drained off safely in order not to pose a risk to persons or property.

5.4.3 Protection against running dry / flow indicator

Warning

An explosive mixture of vapours may form in tanks which are partially empty. The pumps must not run dry in order to prevent explosive atmospheres in storage tanks being transferred to other areas.

The customer should install level indicators in tanks or flow indicators on the discharge side of the pumps to prevent them from running dry.

The intake of air must be prevented.

Caution

Check suction lines regularly to make sure they are tight.

5.4.4 Diaphragm leakage detection

During operation, pumps must not become a source for impermissible medium leaks (this is particularly important where combustible media are dosed).

Caution

The evaluation of the diaphragm leakage signal (electrically isolated reed contact) of the diaphragm leakage detection must be used to stop the pump immediately.

6. Commissioning

For dosing combustible media, see section 4.2.2 *Permissible dosing media*.



Warning

Risk of explosion due to overheating of the pumps.

The pumps must not run against closed valves and fittings.

Warning

The first time the pump is started up or when it is restarted after the dosing head has been emptied, explosive vapour mixtures may form.

The pump must run with a completely full dosing head during start-up and when restarted:

- **Fill the dosing head and suction line with the dosing medium, or,**
- **with flooded suction, deaerate the discharge line to enable the dosing medium to flow into the dosing head.**



7. Operation

7.1 Safety instructions for operating the explosion-proof pump



Warning

Risk of explosion due to overheating of the pumps.

The pumps must not run against closed valves and fittings.



Warning

The pump must not run dry, even during commissioning/recommissioning.

Faults, e.g. reduction in dosing performance, may be caused by damage to the pump, leading to an increased risk of explosion.

Caution

In the event of a fault during operation, the pump must be stopped immediately and the fault corrected.

8. Maintenance



Warning

Risk of explosion due to overheating of the pumps.

Prescribed maintenance operations must be carried out at regular intervals in order to ensure continued conformity with the requirements of Directive 94/9/EC.

8.1 Cleaning

Caution

In order to prevent heat accumulating due to dust deposits on the pump, the pump must be cleaned at regular intervals.

It is the responsibility of the operator to set suitable intervals.
Dust must not be allowed to accumulate > 5 mm.

8.2 Lubrication

Caution

Check the pump regularly for noise generation. The bearings must not overheat.

DMH 25x and DMH 28x: Check the oil level every two weeks. If there is insufficient oil, stop the pump immediately and add oil. Change the oil every 8000 operating hours.
Only use oil approved by Grundfos.

DMX 226: The gear lubricant must be changed every 8000 operating hours by qualified and authorised personnel or by the manufacturer.

8.3 Accessories / peripherals

Danger due to the formation of explosive vapours:

Caution

All seals, screw connections and deaeration screws must be checked regularly to make sure that they are operational and tight.

Check the overflow valve (if installed) regularly to make sure that it is functioning correctly. In the event of a fault in potentially explosive atmospheres, the overflow valve must prevent the gear becoming overloaded and overheated.

Caution

Check the filter in the suction line or on the suction lance regularly, and clean it, if necessary. It is the responsibility of the operator to set suitable intervals.

Caution

Check the lines regularly to make sure that they are tight.

8.4 Tools and maintenance work

Warning

Tools meeting the requirements of EN 1127-1 Appendix A must be used for all operations on pumps in potentially explosive atmospheres. There is a general ban on the use of any type of steel tool in Zone 1 if there is a risk of explosion due to materials in Explosion Group IIC and hydrogen sulphide, ethylene oxide or carbon monoxide.



Warning

Pump maintenance should be carried out in non-explosive atmospheres or only if it can be proven that the area in which repairs are to be carried out is not a potentially explosive atmosphere.



Caution

The original configuration of the pump must not be modified.

For information about spare parts, please refer to the general installation and operating instructions for the pump.

9. Declaration of conformity / certificates

The EC declaration of conformity at the beginning of this booklet replaces the declaration of conformity in the general installation and operating instructions supplied with the pump.

Any modifications made to the pump without our approval will invalidate this declaration.

Before installing the pump, make sure that the system into which it is installed complies with the provisions of the directives. Commissioning is prohibited until this has been established.

The type examination certificate is supplied with the pump on CD (in English and German).

Subject to alterations.

Argentina

Bombas GRUNDFOS de Argentina S.A.
Ruta Panamericana km. 37.500 Lote 34A
1619 - Garin
Pcia. de Buenos Aires
Phone: +54-3327 414 444
Telefax: +54-3327 411 111

Australia

GRUNDFOS Pumps Pty. Ltd.
P.O. Box 2040
Regency Park
South Australia 5942
Phone: +61-8-8461-4611
Telefax: +61-8-8340 0155

Austria

GRUNDFOS Pumpen Vertrieb Ges.m.b.H.
Grundfosstraße 2
A-5082 Grödig/Salzburg
Tel.: +43-6246-883-0
Telefax: +43-6246-883-30

Belgium

N.V. GRUNDFOS Bellux S.A.
Boomssesteenweg 81-83
B-2630 Aartselaar
Tél.: +32-3-870 7300
Télécopie: +32-3-870 7301

Belarus

Представительство ГРУНДФОС в
Минске
220123, Минск,
ул. В. Хоружей, 22, оф. 1105
Тел.: +(375) 17 233 97 65
Факс: (375) 17 233 97 69
E-mail: grundfos_minsk@mail.ru

Bosnia/Herzegovina

GRUNDFOS Sarajevo
Trg Heroja 16,
BiH-71000 Sarajevo
Phone: +387 33 713 290
Telefax: +387 33 659 079
e-mail: grundfos@bih.net.ba

Brazil

BOMBAS GRUNDFOS DO BRASIL
Av. Humberto de Alencar Castelo Branco,
630
CEP 09850 - 300
São Bernardo do Campo - SP
Phone: +55-11 4393 5533
Telefax: +55-11 4343 5015

Bulgaria

Grundfos Bulgaria EOOD
Slatina District
Iztocna Tangenta street no. 100
BG - 1592 Sofia
Tel. +359 2 49 22 200
Fax. +359 2 49 22 201
email: bulgaria@grundfos.bg

Canada

GRUNDFOS Canada Inc.
2941 Brighton Road
Oakville, Ontario
L6H 6C9
Phone: +1-905 829 9533
Telefax: +1-905 829 9512

China

Grundfos Alldos
Dosing & Disinfection
ALLDOS (Shanghai) Water Technology
Co. Ltd.
West Unit, 1 Floor, No. 2 Building (T 4-2)
278 Jinhu Road, Jin Qiao Export Process-
ing Zone
Pudong New Area
Shanghai, 201206
Phone: +86 21 5055 1012
Telefax: +86 21 5032 0596
E-mail: grundfosalldos-CN@grundfos.com

China

GRUNDFOS Pumps (Shanghai) Co. Ltd.
50/F Maxdo Centre No. 8 Xing Yi Rd.
Hongqiao Development Zone
Shanghai 200336
PRC
Phone: +86-21 6122 5222
Telefax: +86-21 6122 5333

Croatia

GRUNDFOS CROATIA d.o.o.
Cebini 37, Buzin
HR-10010 Zagreb
Phone: +385 1 6595 400
Telefax: +385 1 6595 499
www.grundfos.hr

Czech Republic

GRUNDFOS s.r.o.
Čapkovského 21
779 00 Olomouc
Phone: +420-585-716 111
Telefax: +420-585-716 299

Denmark

GRUNDFOS DK A/S
Martin Bachs Vej 3
DK-8850 Bjerringbro
Tlf.: +45-87 50 50 50
Telefax: +45-87 50 51 51
E-mail: info_GDK@grundfos.com
www.grundfos.com/DK

Estonia

GRUNDFOS Pumps Eesti OÜ
Peterburi tee 92G
11415 Tallinn
Tel: + 372 606 1690
Fax: + 372 606 1691

Finland

OY GRUNDFOS Pumpat AB
Mestarintie 11
FIN-01730 Vantaa
Phone: +358-3066 5650
Telefax: +358-3066 56550

France

Pompes GRUNDFOS Distribution S.A.
Parc d'Activités de Chesnes
57, rue de Malacombe
F-38290 St. Quentin Fallavier (Lyon)
Tél.: +33-4 74 82 15 15
Télécopie: +33-4 74 94 10 51

Germany

GRUNDFOS Water Treatment GmbH
Reetzstraße 85
D-76327 Pfinztal (Söllingen)
Tel.: +49 7240 61-0
Telefax: +49 7240 61-177
E-mail: gwt@grundfos.com

Germany

GRUNDFOS GMBH
Schlüterstr. 33
40699 Erkrath
Tel.: +49-(0) 211 929 69-0
Telefax: +49-(0) 211 929 69-3799
E-mail: infoservice@grundfos.de
Service in Deutschland:
E-mail: kundendienst@grundfos.de

Greece

GRUNDFOS Hellas A.E.B.E.
20th km. Athinon-Markopoulou Av.
P.O. Box 71
GR-19002 Peania
Phone: +0030-210-66 83 400
Telefax: +0030-210-66 46 273

Hong Kong

GRUNDFOS Pumps (Hong Kong) Ltd.
Unit 1, Ground floor
Siu Wai Industrial Centre
29-33 Wing Hong Street &
68 King Lam Street, Cheung Sha Wan
Kowloon
Phone: +852-27861706 / 27861741
Telefax: +852-27858664

Hungary

GRUNDFOS Hungária Kft.
Park u. 8
H-2045 Törökbálint,
Phone: +36-23 511 110
Telefax: +36-23 511 111

India

GRUNDFOS Pumps India Private Limited
118 Old Mahabalipuram Road
Thoraiakkam
Chennai 600 097
Phone: +91-44 4596 6800

Indonesia

PT GRUNDFOS Pompa
Jl. Rawa Sumur III, Blok III / CC-1
Kawasan Industri, Pulogadung
Jakarta 13930
Phone: +62-21-460 6909
Telefax: +62-21-460 6910 / 460 6901

Ireland

GRUNDFOS (Ireland) Ltd.
Unit A, Merrywell Business Park
Ballymount Road Lower
Dublin 12
Phone: +353-1-4089 800
Telefax: +353-1-4089 830

Italy

GRUNDFOS Pompe Italia S.r.l.
Via Gran Sasso 4
I-20060 Truccazzano (Milano)
Tel.: +39-02-95838112
Telefax: +39-02-95309290 / 95838461

Japan

GRUNDFOS Pumps K.K.
Gotanda Metalion Bldg. 5F,
5-21-15, Higashi-gotanda
Shiagawa-ku, Tokyo,
141-0022 Japan
Phone: +81 35 448 1391
Telefax: +81 35 448 9619

Korea

GRUNDFOS Pumps Korea Ltd.
6th Floor, Aju Building 679-5
Yeoksam-dong, Kangnam-ku, 135-916
Seoul, Korea
Phone: +82-2-5317 600
Telefax: +82-2-5633 725

Latvia

SIA GRUNDFOS Pumps Latvia
Deglava biznesa centrs
Augusta Deglava ielā 60, LV-1035, Rīga,
Tālr.: + 371 714 9640, 7 149 641
Fakss: + 371 914 9646

Lithuania

GRUNDFOS Pumps UAB
Smolensko g. 6
LT-03201 Vilnius
Tel: + 370 52 395 430
Fax: + 370 52 395 431

Malaysia

GRUNDFOS Pumps Sdn. Bhd.
7 Jalan Peguam U1/25
Glenmarie Industrial Park
40150 Shah Alam
Selangor
Phone: +60-3-5569 2922
Telefax: +60-3-5569 2866

Mexico

Bombas GRUNDFOS de México S.A. de
C.V.
Boulevard TLC No. 15
Parque Industrial Stiva Aeropuerto
Apodaca, N.L. 66600
Phone: +52-81-8144 4000
Telefax: +52-81-8144 4010

Netherlands

GRUNDFOS Netherlands
Veluwezoom 35
1326 AE Almere
Postbus 22015
1302 CA ALMERE
Tel.: +31-88-478 6336
Telefax: +31-88-478 6332
E-mail: info_gnl@grundfos.com

New Zealand

GRUNDFOS Pumps NZ Ltd.
17 Beatrice Tinsley Crescent
North Harbour Industrial Estate
Albany, Auckland
Phone: +64-9-415 3240
Telefax: +64-9-415 3250

Norway

GRUNDFOS Pumper A/S
Strømsveien 344
Postboks 235, Leirdal
N-1011 Oslo
Tlf.: +47-22 90 47 00
Telefax: +47-22 32 21 50

Poland

GRUNDFOS Pompy Sp. z o.o.
ul. Klonowa 23
Baranowo k. Poznania
PL-62-081 Przeźmierowo
Tel: (+48-61) 650 13 00
Fax: (+48-61) 650 13 50

Portugal

Bombas GRUNDFOS Portugal, S.A.
Rua Calvet de Magalhães, 241
Apartado 1079
P-2770-153 Paço de Arcos
Tel.: +351-21-440 76 00
Telefax: +351-21-440 76 90

Romania

GRUNDFOS Pompe România SRL
Bd. Biruintei, nr 103
Pantelimon county Ilfov
Phone: +40 21 200 4100
Telefax: +40 21 200 4101
E-mail: romanian@grundfos.ro

Russia

ООО Грундфос
Россия, 109544 Москва, ул. Школьная
39
Тел. (+7) 495 737 30 00, 564 88 00
Факс (+7) 495 737 75 36, 564 88 11
E-mail grundfos.moscow@grundfos.com

Serbia

GRUNDFOS Predstavništvo Beograd
Dr. Milutina Ivkovića 2a/29
YU-11000 Beograd
Phone: +381 11 26 47 877 / 11 26 47 496
Telefax: +381 11 26 48 340

Singapore

GRUNDFOS (Singapore) Pte. Ltd.
24 Tuas West Road
Jurong Town
Singapore 638381
Phone: +65-6865 1222
Telefax: +65-6861 8402

Slovenia

GRUNDFOS d.o.o.
Šlandrova 8b, SI-1231 Ljubljana-Črnuče
Phone: +386 1 568 0610
Telefax: +386 1 568 0619
E-mail: slovenia@grundfos.si

South Africa

Grundfos (PTY) Ltd.
Corner Mountjoy and George Allen Roads
Wilbart Ext. 2
Bedfordview 2008
Phone: (+27) 11 579 4800
Fax: (+27) 11 455 6066
E-mail: lsmart@grundfos.com

Spain

Bombas GRUNDFOS España S.A.
Camino de la Fuentecilla, s/n
E-28110 Algete (Madrid)
Tel.: +34-91-848 8800
Telefax: +34-91-628 0465

Sweden

GRUNDFOS AB
(Box 333) Lunnagårdsgatan 6
431 24 Mölndal
Tel.: +46 31 332 23 000
Telefax: +46 31-331 94 60

Switzerland

GRUNDFOS ALLDOS International AG
Schönmattpstraße 4
CH-4153 Reinach
Tel.: +41-61-717 5555
Telefax: +41-61-717 5500
E-mail: grundfosalldos-CH@grundfos.com

Switzerland

GRUNDFOS Pumpen AG
Bruggacherstrasse 10
CH-8117 Fällanden/ZH
Tel.: +41-1-806 8111
Telefax: +41-1-806 8115

Taiwan

GRUNDFOS Pumps (Taiwan) Ltd.
7 Floor, 219 Min-Chuan Road
Taichung, Taiwan, R.O.C.
Phone: +886-4-2305 0868
Telefax: +886-4-2305 0878

Thailand

GRUNDFOS (Thailand) Ltd.
92 Chaloe Phrakiat Rama 9 Road,
Dokmai, Pravej, Bangkok 10250
Phone: +66-2-725 8999
Telefax: +66-2-725 8998

Turkey

GRUNDFOS POMPA San. ve Tic. Ltd. Sti.
Gebze Organize Sanayi Bölgesi
İhsan dede Caddesi,
2. yol 200. Sokak No. 204
41490 Gebze/ Kocaeli
Phone: +90 - 262-679 7979
Telefax: +90 - 262-679 7905
E-mail: satis@grundfos.com

Ukraine

ТОВ ГРУНДФОС УКРАЇНА
01010 Київ, Вул. Московська 86,
Тел.: (+38 044) 390 40 50
Факс.: (+38 044) 390 40 59
E-mail: ukraine@grundfos.com

United Arab Emirates

GRUNDFOS Gulf Distribution
P.O. Box 16768
Jebel Ali Free Zone
Dubai
Phone: +971-4- 8815 166
Telefax: +971-4-8815 136

United Kingdom

GRUNDFOS Pumps Ltd.
Grovebury Road
Leighton Buzzard/Beds. LU7 8TL
Phone: +44-1525-850000
Telefax: +44-1525-850011

U.S.A.

GRUNDFOS Pumps Corporation
17100 West 118th Terrace
Olathe, Kansas 66061
Phone: +1-913-227-3400
Telefax: +1-913-227-3500

Uzbekistan

Представительство ГРУНДФОС в
Ташкенте
700000 Ташкент ул.Усмана Носира 1-й
тулик 5
Телефон: (3712) 55-68-15
Факс: (3712) 53-36-35

Addresses revised 27.04.2012

96681420 0712
ECM: 1085243

The name Grundfos, the Grundfos logo, and the payoff Be-Think-Innovate are registered trademarks owned by Grundfos Management A/S or Grundfos A/S, Denmark. All rights reserved worldwide.